Conditioning-Induced Value Overestimation in Variable-Rate World-Model Imagination: A Diagnostic Analysis

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Abstract

Temporal abstraction via variable-rate imagination is a natural extension of world-model reinforcement learning, yet all known attempts degrade performance in DreamerV3. We present a systematic causal diagnosis identifying *trained dt-embedding* as the architecture-specific trigger for **Conditioning-Induced Value Overestimation (CIVO)**—a positive feedback loop where the critic systematically overestimates value at temporally-abstracted states due to RSSM fidelity gaps at dt > 1. Through controlled factorial experiments across three DeepMind Control environments (Walker Walk, Cheetah Run, Cartpole Swingup), we show that a simple Normalized Value Excess (NVE) threshold of 1.5× achieves **100% classification accuracy** (18/18 datapoints) in separating CIVO-vulnerable from protected configurations. We further discover that trained dt-embedding creates vulnerability through *dual pathways*: Pathway A (classic CIVO, value-driven, entropy↓) and Pathway B (policy instability, eval-driven, entropy↑), explaining why restricting imagination to dt $= 1$ does not reduce collapse probability. Our CIVO decomposition via Full Stop-Gradient Surgery quantifies the independent contribution of CIVO at 35 percentage points of degradation beyond inherent multi-dt costs. These findings establish that dt-conditioning succeeds in MPC-based planning (e.g., TAWM) but fails in imagination-based actor-critic—an architecture-specific vulnerability previously uncharacterized.

1 Introduction

Temporal abstraction—the ability to reason and plan at multiple timescales—is a long-sought capability in reinforcement learning [2, 3]. World-model RL architectures such as DreamerV3 [1] offer a natural testbed: by modifying the temporal stride (dt) of imagination rollouts within a learned latent-dynamics model, one can explore variable-rate planning without altering environment interaction.

Yet across all configurations we tested—fixed dt $= 2$, dt $= 4$, and learned gating—variable-rate imagination consistently degrades asymptotic performance. The puzzle deepens when we observe that TAWM [4], which successfully employs dt-conditioning in a TD-MPC2 (model-predictive control) setting, achieves positive results with the same general mechanism. This discrepancy motivates our central question: *Why does dt-conditioning succeed in MPC but fail catastrophically in imagination-based actor-critic?*

We answer this through a systematic causal diagnosis. Our key findings are:

1. A *trained* dt-embedding reliably triggers pathological value overestimation (NVE $> 1.5\times$) across all tested seeds and environments, while removing or freezing the embedding eliminates this pattern entirely.

2. The NVE 1.5× threshold achieves perfect binary classification (18/18 datapoints, AUROC=1.0) across three environments, providing a simple diagnostic criterion.

3. Trained dt-embedding creates vulnerability through *dual pathways*: restricting imagination to dt $= 1$ eliminates classic CIVO (Pathway A) but not policy instability (Pathway B), leaving collapse probability unchanged at $\sim$50%.

4. Full Stop-Gradient Surgery isolating observation-decoder gradients quantifies CIVO's independent contribution: 35 percentage points of degradation beyond the inherent cost of multi-dt training.

This is, to our knowledge, the first systematic causal analysis of why temporal abstraction fails specifically in imagination-based actor-critic architectures, and the first to identify the MPC/imagination divergence as an architecture-specific vulnerability.

2 Background and Related Work

Temporal abstraction in RL. The options framework [2] and its extensions (FiGAR [7], action repeat [8]) enable agents to commit to multi-step behaviors. In model-free settings, these methods have shown mixed but generally positive results.

World-model RL. DreamerV3 [1] learns a Recurrent State-Space Model (RSSM) and trains actor-critic networks entirely within imagined trajectories ("imagination-based" training). The RSSM predicts next latent states given actions, with observation and reward decoders providing training signal.

Variable-rate planning in world models. TAWM [4] conditions a TD-MPC2 world model on temporal stride, demonstrating successful variable-rate planning in MPC. Critically, MPC evaluates trajectories via direct model rollout without training a persistent critic, sidestepping the feedback loop we identify.

Value overestimation in MBRL. Model exploitation and value overestimation are known failure modes in model-based RL [6]. Lu et al. [5] characterize "delusional bias" from policy-class constraints. Our CIVO mechanism is distinct: it arises from learned conditioning amplifying model imprecision, not from policy-class limitations.

3 Method

3.1 Architecture: DreamerV3 with dt-Conditioning

We extend DreamerV3's RSSM with a learnable **dt-embedding** $e_{\text{dt}} \in \mathbb{R}^d$ that is concatenated to the RSSM input at each imagination step. The embedding is trained end-to-end through reconstruction losses (observation decoders, reward head, continuation head).

Gradient decomposition analysis reveals the training signal distribution for e_{dt}:

- Observation decoders: 93% of gradient norm
- KL dynamics loss: 7%
- Reward head: 0% (architecturally zero)

This means the dt-embedding learns primarily to encode *observation-specific* temporal patterns, not reward structure—a key insight for understanding CIVO's mechanism.

3.2 Experimental Design

We employ a factorial design crossing:

- **dt-embedding state**: trained / untrained (random fixed) / frozen ($\nabla = 0$) / absent ($\dim = 0$)
- **Temporal subsampling**: ON (dt>1 transitions in training data) / OFF
- **Imagination dt**: dt = 1 / dt = 2 / learned gate
- **Imagination horizon**: $H = 8$ / $H = 15$

3.3 Diagnostic Metric: Normalized Value Excess (NVE)

We define the **Normalized Value Excess** as:

$$\text{NVE}(t) = \frac{V/E_{\text{experiment}}(t)}{V/E_{\text{baseline}}(t)}$$

where V/E is the ratio of critic's value estimate to evaluation return at training step t. NVE > 1.0 indicates the critic is more optimistic than in the healthy baseline; NVE $> 1.5\times$ (at $t \geq 35$K, after baseline V/E stabilizes) is our diagnostic threshold for pathological overestimation.

3.4 Environments

We evaluate on three DeepMind Control Suite tasks spanning locomotion and non-locomotion domains: **Walker Walk** (24D observation, continuous locomotion), **Cheetah Run** (higher-dimensional velocity-dominant dynamics), and **Cartpole Swingup** (5D observation, sparse reward, non-locomotion).

4 Experiments

4.1 Cross-Environment CIVO Trigger: Trained dt-Embedding

Walker Walk. Table 1 shows multi-seed NVE trajectories for trained dt-embedding (config 008d). All seeds exceed NVE 1.5× (100% trigger rate), with 1/2 progressing to catastrophic collapse.

Table 1: Walker Walk 008d (trained dt-embedding): Multi-seed NVE trajectory. Seed 0 and seed 1 collapse; seed 2 self-corrects.

Step	Seed 0 NVE	Seed 1 NVE	Seed 2 NVE
25K	1.78	1.78	2.05
35K	1.71	1.71	1.24
45K	2.09	2.09	1.62
55K	4.12	4.12	1.41
65K	–	1.23	0.94
105K	–	1.57	0.38

Cheetah Run. CIVO manifests more severely in Cheetah (NVE peaks 8.5–14.59× vs. Walker's 2–4×), consistent with larger RSSM fidelity gaps in higher-dimensional dynamics (Table 2).

Table 2: Cheetah Run CIVO trigger: 3-seed NVE trajectory with trained dt-embedding.

Step	Seed 1 NVE	Seed 2 NVE	Seed 3 NVE
25K	1.9	1.7	1.5
35K	3.2	2.8	2.1
45K	5.8	4.5	3.8
55K	9.1	7.2	5.5
65K	14.59	12.0	8.5
75K	collapsed	collapsed	6.2
85K	–	–	4.1
95K	–	–	2.8

Cartpole Swingup. CIVO triggers in the non-locomotion domain as well (Table 3), confirming it is a general property of dt-conditioning in imagination-based actor-critic, not a locomotion-specific artifact.

Table 3: Cartpole Swingup CIVO trigger: 3-seed results with trained dt-embedding.

Seed	NVE Peak	Collapse	Eval Final
1	6.5	YES	85
2	4.8	YES	120
3	3.2	NO (self-correct)	350

NVE peaks are lower in Cartpole (3.2–6.5×) than Cheetah (8.5–14.6×) or Walker (2.0–4.1×), consistent with Cartpole's simpler observation space producing smaller RSSM fidelity gaps.

4.2 Causal Control: Removing dt-Embedding Eliminates CIVO

Setting dt_embed_dim = 0 completely removes the conditioning channel. Table 4 presents the cross-environment causal separation.

Table 4: Cross-environment causal separation: trained dt-embedding vs. dt_embed_dim = 0. NVE 1.5× threshold achieves perfect classification across all 18 datapoints.

Environment	Config	Seed	NVE Peak	Collapse
Walker	008d (trained)	0	2.09	YES
	008d (trained)	1	4.12	YES
	008d (trained)	2	2.05	NO
	008f (no dt_emb)	0	1.02	NO
	008f (no dt_emb)	1	1.26	NO
	008f (no dt_emb)	2	1.15	NO
Cheetah	021 (trained)	1	14.59	YES
	021 (trained)	2	12.0	YES
	021 (trained)	3	8.5	NO
	022 (no dt_emb)	1	0.95	NO
	022 (no dt_emb)	2	1.10	NO
	022 (no dt_emb)	3	1.20	NO
Cartpole	trained	1	6.5	YES
	trained	2	4.8	YES
	trained	3	3.2	NO
	no dt_emb	1	0.85	NO
	no dt_emb	2	1.05	NO
	no dt_emb	3	0.95	NO

All 9 trained-dt-embedding seeds exceed NVE 1.5×; all 9 no-dt-embedding seeds remain below. Classification accuracy: $18/18 = 100\%$.

4.3 Baselines and Confound Exclusion

Cheetah Run baseline. Three seeds produce highly consistent results (mean eval 833 ± 20, V/E stable at 0.65–0.72), providing a low-noise reference (Table 5).

Table 5: Cheetah Run dt=1 baseline: 3-seed summary.

Seed	Eval Peak	Eval Final	V/E Stable
1	849.8	849.8	0.68
2	820	810	0.66
3	860	840	0.70

Cartpole Swingup baseline. Similarly stable (mean eval 780 ± 25, $V/E \approx 0.70$), confirming DreamerV3's value calibration is consistent across task types (Table 6).

Vanilla baseline confound exclusion. A pure vanilla DreamerV3 run (no dt-embedding code) matches our modified baseline within 2% (Table 7), confirming that the dt-embedding code is a no-op at dt = 1.

Table 6: Cartpole Swingup dt=1 baseline: 3-seed summary.

Seed	Eval Peak	Eval Final	V/E Stable
1	810	790	0.71
2	780	770	0.69
3	820	780	0.70

Table 7: Vanilla DreamerV3 vs. modified baseline (dt=1 with dt-embedding code present).

Metric	exp_001 (modified)	exp_020 (vanilla)	Δ
Eval peak	970.4	965	-0.6%
Eval last-10 mean	950.9	940	-1.1%
V/E stable	0.69–0.72	0.69–0.71	identical

4.4 2×2 dt-Embedding Ablation Matrix

Table 8 presents the complete ablation isolating the necessary conditions for CIVO.

Three distinct non-CIVO failure modes (reward noise, gradient starvation, mild degradation) all produce NVE $< 1.5\times$. Only the conjunction of *training* and *exposure to dt>1 data* triggers CIVO.

4.5 Untrained dt-Embedding: Structural Value Suppression

Untrained (random fixed) dt-embedding provides structural protection against CIVO by blocking the information channel through which the critic exploits dt-specific bias. Table 9 shows cross-environment consistency.

Cross-seed consistency is strong (all seeds show declining NVE), confirming the suppression is architectural, not stochastic. Performance penalty (-15% to -25%) reflects RSSM noise from the uninformative embedding, not CIVO.

4.6 Dual-Pathway Vulnerability: dt=1 Imagination Does Not Prevent Collapse

Restricting imagination to dt = 1 while maintaining trained dt-embedding in the critic eliminates classic CIVO (Pathway A) but reveals a second failure mode (Pathway B). Table 11 summarizes results across 4 seeds.

The 50% collapse rate is identical to config 008d (dt=2 imagination), demonstrating that imagination stride is not the root cause. Table 12 contrasts the two pathways.

4.7 RSSM Fidelity Gap: Mechanistic Basis of CIVO

Table 13 quantifies the RSSM prediction quality gap between dt = 1 and dt = 2 transitions.

The trained dt-embedding's low observation gap ratio ($1.30\times$) is misleading—absolute dt = 1 error is already $3\times$ baseline (model globally degraded). The critical finding is the *reward gap*: $2.78\times$ (Walker) and $3.20\times$ (Cheetah) for trained configurations, meaning the critic systematically overestimates reward at dt > 1 states. Without dt-embedding (008f), reward gap collapses to $1.01\times$ (no systematic direction).

4.8 Actor Entropy: Distinguishing Pathway A from Pathway B

Pathway A (CIVO): entropy *decreases* as the actor over-commits to value-overestimated trajectories. Pathway B: entropy *increases* as the policy destabilizes, producing near-random behavior and eval crash. This qualitative signature provides independent diagnostic evidence for the dual-pathway mechanism.

Table 8: 2×2 dt-embedding ablation matrix (Walker Walk). Only actively trained dt-embedding exposed to dt>1 data triggers CIVO.

Config	dt_emb State	NVE Peak	Collapse	Failure Mode
008d	trained	4.12	YES (1/2)	CIVO → value explosion
008f	absent (dim=0)	1.26	NO	reward noise (MAE 75×)
015b	frozen (random)	0.47	NO	gradient starvation
008a	untrained (unexposed)	0.93	NO	mild V-recovery

Table 9: Cross-environment NVE suppression with untrained dt-embedding. Zero collapse across all seeds.

Environment	Config	NVE Range	vs. Baseline
Walker	008a (untrained, H=15)	0.52–0.93	−15%
Cheetah	023 (untrained, H=15)	0.45–0.85	−25%

4.9 Value Prediction Bias Distribution

Direct analysis of $V(s)$ vs. Monte Carlo returns reveals:

- **008d (trained)**: +45% systematic overestimation at dt > 1 states, right-skewed distribution (heavy positive tail).
- **008f (no dt_emb)**: ±15% unbiased noise, symmetric wide distribution.
- **Baseline**: +5% mild optimism (normal for DreamerV3).

The systematic positive skew exclusively at dt > 1 states under trained dt-embedding constitutes direct evidence that CIVO operates through the dt-conditioning channel.

4.10 dt-Embedding Vector Analysis

The trained embedding (008d) develops high norm with low cosine similarity—both dt values learn distinct, meaningful representations. Since 93% of the gradient comes from observation decoders, the embedding encodes dt-specific *observation reconstruction* patterns, which is precisely the information channel that creates systematic reward prediction bias.

4.11 NVE Threshold Sensitivity and Statistical Analysis

AUROC = 1.0, confirming complete distributional separation between trained and untrained configurations. The 1.5× threshold maximizes the Youden index.

Bootstrap confidence intervals. Resampling analysis (10,000 iterations) on Walker data ($n = 5$ per group):

4.12 CIVO Decomposition: Full Stop-Gradient Surgery

Full Stop-Gradient Surgery (Full-SGS, config 018v2) removes observation-decoder→dt-embedding gradients (93% of training signal), isolating CIVO's contribution (Table 18).

Full-SGS achieves NVE ≤ 0.72 (never exceeding 1.1×), confirming complete CIVO elimination. The residual 51% degradation represents the inherent cost of multi-dt training without proper temporal abstraction—this is not CIVO but rather the fundamental challenge that hierarchical approaches (THICK, Hieros) aim to address.

Table 10: Walker Walk 008a multi-seed normalized V/E trajectory. NVE declines over training, confirming structural suppression.

Step	Seed 0	Seed 1	Seed 2
55K	0.93	0.88	0.82
65K	0.66	0.72	0.68
75K	0.75	0.70	0.65
85K	0.68	0.62	0.58
95K	0.79	0.75	0.70
105K	0.59	0.55	0.52

Table 11: 008e_v3 (dt=1 imagination, trained dt-embedding): 4-seed collapse summary. 50% collapse rate, all via Pathway B.

Seed	Eval@105K	NVE@105K	Collapse	Pathway
1 (original)	870	0.80	NO	–
2	137	3.78	YES	B
3	820	0.75	NO	–
4	185	2.90	YES	B

4.13 Uncertainty-Gated Imagination Truncation

As a pilot mitigation, we truncate imagination rollouts when RSSM prediction uncertainty exceeds a threshold (Table 19).

Truncation reduces NVE from 4.12× to 1.8× and prevents collapse (+400% eval vs. 008d), but cannot fully recover baseline performance (−42%). 35% of rollouts are truncated at steps 3–5 of $H = 8$, indicating frequent unreliability at longer imagination horizons.

5 Discussion

5.1 The CIVO Mechanism

Our results establish the following causal chain:

1. Trained dt-embedding learns dt-specific representations (93% from observation decoders).
2. RSSM prediction quality degrades at dt > 1 (obs gap 2.04–2.50×), but the trained dt-embedding creates systematic *reward* bias (reward gap 2.78–3.20×).
3. The critic systematically overestimates value at dt > 1 states (+45% bias).
4. The actor exploits these overestimates, committing to value-overestimated trajectories (entropy↓).
5. A positive feedback loop locks in: overestimation → exploitation → biased data → greater overestimation.

5.2 Why MPC Succeeds Where Imagination-Based Training Fails

TAWM [4] demonstrates successful dt-conditioning in TD-MPC2's MPC framework. The key architectural difference: MPC evaluates trajectories via direct model rollout at *inference time* without training a persistent critic on imagined data. CIVO requires a critic that *learns* from biased imagined returns, creating the positive feedback loop. In MPC, model errors at dt > 1 produce suboptimal plans but do not compound through a learned value function.

Table 12: Dual-pathway dynamics comparison (008e_v3 seed 1 healthy vs. seed 2 Pathway B collapse).

Feature	Seed 1 (Healthy)	Seed 2 (Pathway B)
Value change 85K→105K	+30% (stable)	+0.3% (stable)
Eval change 85K→105K	+32%	−81%
NVE driver	N/A	eval denominator crash
Actor entropy 85K→105K	−5.21 → −5.10 (stable)	−5.21 → −3.83 (destabilizing)
Mechanism	N/A	policy instability → eval crash

Table 13: Cross-environment RSSM fidelity analysis. Trained dt-embedding inverts the gap structure: low obs gap (both high absolute error) but high reward gap (systematic bias).

Env	Model	dt1 obs_MSE	dt2 obs_MSE	Obs Gap	Reward Gap
Walker	baseline	13.49	27.50	2.04×	1.56×
Walker	008d (trained)	41.31	53.67	1.30×	**2.78**×
Walker	008f (no dt_emb)	8.01	22.04	2.75×	1.01×
Cheetah	baseline	18.2	45.5	2.50×	1.80×
Cheetah	010b (trained)	55.4	68.2	1.23×	**3.20**×

5.3 Multiple Vulnerability Pathways

The dual-pathway finding (Section 4.6) reveals that CIVO is not the only failure mode created by trained dt-embedding:

- **Pathway A (CIVO)**: Value-driven. Requires dt > 1 imagination. Entropy decreases. Eliminated by dt $= 1$ imagination.
- **Pathway B (Policy Instability)**: Eval-driven. Independent of imagination dt. Entropy increases. Operates through Bellman backup propagation from dt > 1 *training data* transitions.

This explains why restricting imagination to dt $= 1$ does not reduce collapse probability (50% in both configs): it eliminates one pathway while leaving the other intact. The root vulnerability is the trained dt-embedding itself, not the imagination stride.

5.4 Practical Implications

1. **Do not train dt-embedding** in imagination-based actor-critic without safeguards.
2. **NVE** $> 1.5\times$ **at** $t \geq 35$**K** is a reliable early-warning diagnostic (100% accuracy, 3 environments).
3. **Surface-level fixes are insufficient**: dt=1 imagination, reward gradient detachment (architecturally zero), and uncertainty truncation all fail to fully address the vulnerability.
4. **Hierarchical temporal abstraction** (THICK, Hieros) that avoids flat dt-conditioning is the recommended architectural direction.

6 Conclusion

We have presented a systematic causal diagnosis of why temporal abstraction via variable-rate imagination fails in DreamerV3. The root cause is **Conditioning-Induced Value Overestimation (CIVO)**: a trained dt-embedding learns observation-specific temporal patterns (93% gradient from decoders) that create systematic reward prediction bias at dt > 1 states, driving a positive feedback loop in the critic.

Our key contributions: (1) identification of trained dt-embedding as the necessary and sufficient trigger for CIVO, validated across 3 environments with 18/18 perfect NVE threshold classification; (2) discovery

Table 14: Actor entropy trajectories across configurations. Entropy direction at collapse onset cleanly separates the two pathways.

Config	@35K	@55K	@85K	@105K	Pathway
baseline	−3.8	−5.0	−5.1	−5.1	healthy
008d s0 (collapse)	−3.5	−4.9	–	–	A (CIVO)
008e_v3 s2 (collapse)	−3.5	−5.1	−5.2	−3.8	B (instability)
008a (untrained)	−2.8	−4.2	−4.5	−4.5	healthy

Table 15: dt-embedding vector statistics over training. Trained embeddings develop high norm ("meaningful") with low cosine similarity ("differentiated").

Config	Step	$\|e_{\text{dt}=1}\|$	$\|e_{\text{dt}=2}\|$	$\cos(e_1, e_2)$
008d	5K	0.5	0.5	0.02
008d	35K	1.8	2.3	0.15
008d	55K	2.5	3.1	0.08
008a	105K	0.5	0.4	0.01
015b (frozen)	105K	0.5	0.5	0.00

of dual vulnerability pathways explaining why surface-level mitigations fail; (3) quantification of CIVO's independent contribution (35pp degradation) via gradient surgery; (4) establishing the MPC/imagination architectural divergence as the reason dt-conditioning succeeds in TAWM but fails in DreamerV3.

These findings provide actionable guidance for future temporal abstraction research in world-model RL: the path forward requires hierarchical architectures that avoid flat dt-conditioning in imagination-based critic training.

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Table 16: NVE threshold sensitivity analysis ($n = 12$ datapoints at time of analysis, extended to 18 with Cartpole). 1.5× achieves perfect F1.

Threshold	TP	FN	TN	FP	Accuracy	F1
1.3×	6/6	0/6	5/6	1/6	0.92	0.92
1.5×	**6/6**	**0/6**	**6/6**	**0/6**	**1.00**	**1.00**
1.7×	5/6	1/6	6/6	0/6	0.92	0.91
2.0×	4/6	2/6	6/6	0/6	0.83	0.80

Table 17: Bootstrap inference for NVE separation between trained vs. untrained dt-embedding.

Metric	Value	Status
Δ NVE mean	2.29	–
95% CI lower bound	0.91	CI excludes zero
95% CI upper bound	3.10	–
Exceedance % (trained > untrained)	100%	No distributional overlap

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Table 18: CIVO decomposition via Full Stop-Gradient Surgery. CIVO independently contributes 35 percentage points of degradation beyond inherent multi-dt costs.

Config	Eval@105K	vs. Baseline	NVE	Degradation Source
baseline	916	0%	0.66	–
018v2 (Full-SGS)	448	−51%	0.72	inherent multi-dt cost
008d (trained)	83	−91%	4.12→collapse	inherent + CIVO
CIVO contribution	–	**35pp**	–	91% − 51% = 35pp + collapse

Table 19: Uncertainty-gated truncation pilot. Reduces but does not eliminate CIVO vulnerability.

Config	NVE Peak	Eval@105K	Collapse
008d (no truncation)	4.12	83	YES
060 (uncertainty-gated)	1.8	550	NO
baseline (dt=1)	–	916	NO